Yagoub Rabich

CMAN|CF07

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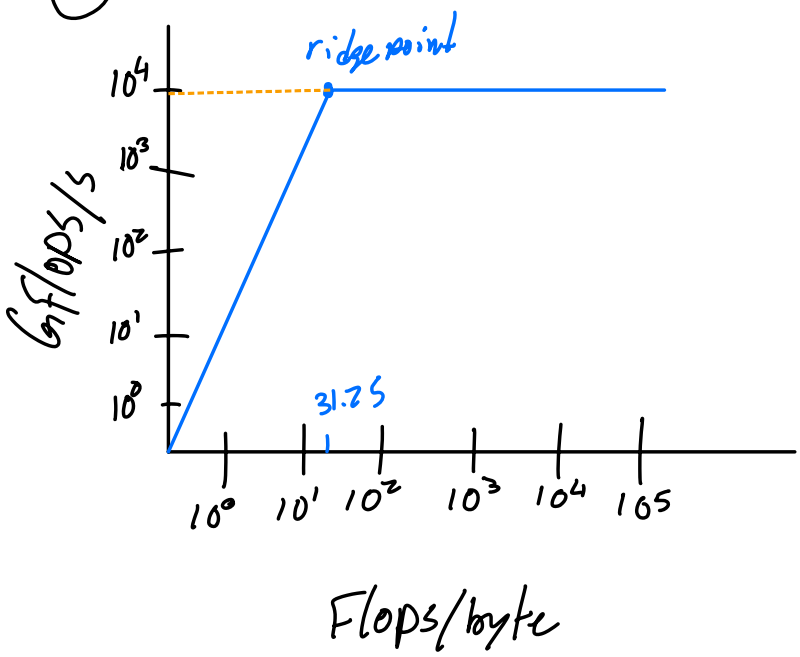
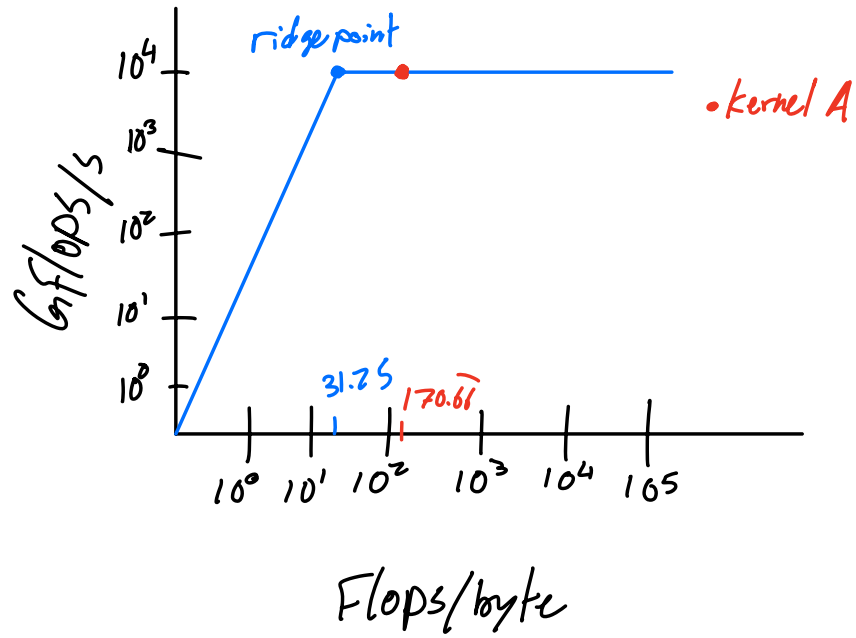


2 $N = 1024$

$1024 \times 1024 = 1048576$ elements $\times$ 3 matrices

$= 3145728 \times 4$ bytes $= 12582912$ bytes

$$2 \times 1024^3 = \frac{2147483684 \text{ flops}}{12582912 \text{ bytes}} = \frac{512}{3} = 170.\overline{666}$$



③

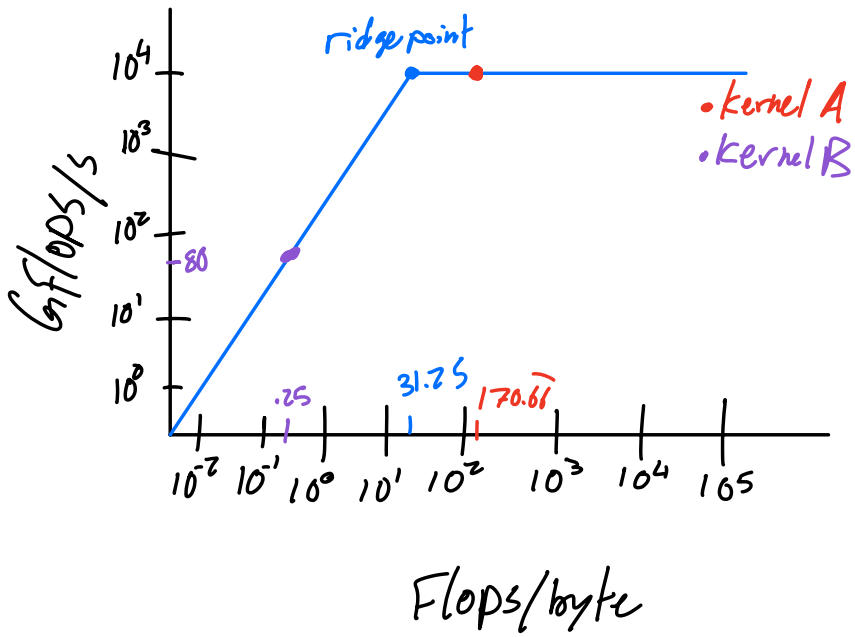
each element requires 1 addition

so, Flops = 4194304

4194304 x 4 bytes = 16777216 bytes

$$\frac{4194304}{16777216} = \frac{1}{4} \text{ flops/byte}$$

$$\frac{1}{4} \times 320 \text{ GB/s} = 80 \text{ GFlops/s}$$



(4)

(a)

Kernel A is Compute bound
kernel B is memory-bandwidth bound

(B)

Attainable performance ceiling for kernel A is 10k GFlops/s
Attainable performance ceiling for kernel B is 80 GFlops/s

(C)

Kernel A needs more compute power like more Flops/s, more cores, or higher clock speeds
Kernel B needs more memory bandwidth. so a wider memory bus or an HBM